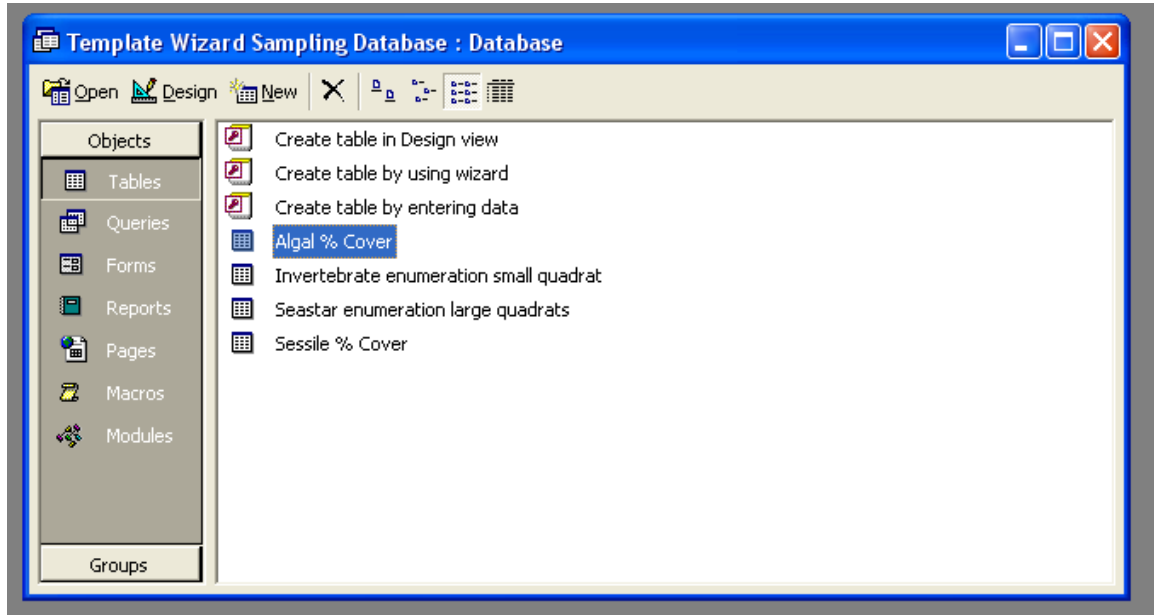


Appendix - Access Methods

All of the Wizard Island data from 1997-2009 (as well as data collected by the expansion group in 2009 at other sites) has been entered into a Microsoft Access database. Data can be moved between Access and Excel quite easily. Access allows researchers to subset and analyze data in more concise ways, but has its limitations where Excel may be more appropriate.

When Access is opened you will immediately get this screen:



There are 3 “Objects” that are important for our purposes: Tables, Queries, and Forms.

Tables	Contain all of the data in a spreadsheet format. Data is rarely ever manipulated within the table itself.
Queries	Allow sub-sampling and manipulation of the tables.
Forms	Used for inputting data. They are linked directly to the table where they deposit the data. They help prevent data entry error.

Design View of a table



Microsoft Access

File Edit View Insert Format Records Tools Window Help Adobe PDF

Algal % Cover - Table

Date	Site	Exposure	Transect #	Tidal Height	Acrosiphonia spp	Ahnfeltia pacifica	Alaria marginata	Analipus japei	Chondracanthi	Cladophora sp	Codium fragile	Copamar
02-Aug-97	Wizard	Exposed	1	0.5	0	0	1	0	0	0	0	0
02-Aug-97	Wizard	Exposed	1	1	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	1	1.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	1	2	0	0	0	0	0	5	0	0
02-Aug-97	Wizard	Exposed	1	2.5	0	0	0	0	0	5	0	0
02-Aug-97	Wizard	Exposed	1	3	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	1	3.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	10	0.5	0	0	0	0	5	0	0	0
02-Aug-97	Wizard	Exposed	10	1	0	0	6	0	0	0	0	0.5
02-Aug-97	Wizard	Exposed	10	1.5	0	0	0	0	0	0	10	0
02-Aug-97	Wizard	Exposed	10	2	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	10	2.5	0	0	3	0	0	0	45	0
02-Aug-97	Wizard	Exposed	10	3	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	10	3.5	0	0	0	0	0	0	35	0
02-Aug-97	Wizard	Exposed	11	0.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	11	1	0	0	5	0	0	0	0	0
02-Aug-97	Wizard	Exposed	11	1.5	0	0	0	0	0	0	25	0
02-Aug-97	Wizard	Exposed	11	2	0	0	0	0	0	0	25	0
02-Aug-97	Wizard	Exposed	11	2.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	11	3	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	11	3.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	12	0.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	12	1	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	12	1.5	0	0	0	0	0	0	10	0
02-Aug-97	Wizard	Exposed	12	2	0	0	0	0	0	0	0.5	0
02-Aug-97	Wizard	Exposed	12	2.5	0	0	0	0	0	0	30	0
02-Aug-97	Wizard	Exposed	12	3	0	0	0	0	0	0	20	0
02-Aug-97	Wizard	Exposed	12	3.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	13	0.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	13	1	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	13	1.5	0	0	0	0	0	0	15	0
02-Aug-97	Wizard	Exposed	13	2	0	0	0	0	0	0	40	0
02-Aug-97	Wizard	Exposed	13	2.5	0	0	0	0	0	0	0.5	0
02-Aug-97	Wizard	Exposed	13	3	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	13	3.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	14	0.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	14	1	0	0	65	1	0	0	0	0
02-Aug-97	Wizard	Exposed	14	1.5	0	0	0	0	0	0	15	0
02-Aug-97	Wizard	Exposed	14	2	0	0	0	0	0	0	40	0
02-Aug-97	Wizard	Exposed	14	2.5	0	0	0	0	0	0	0.5	0
02-Aug-97	Wizard	Exposed	14	3	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	14	3.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	15	0.5	0	0	0	0	0	0	0	0
02-Aug-97	Wizard	Exposed	15	1	0	0	0	0	0	0	0	0

Record: 14 of 1000

From: Harrington et al. 2009. Long term monitoring of the rocky intertidal on Wizard Island in Barkley Sound. Coastal Biodiversity and Conservation Student Report, no. 330. Bamfield, BC: Bamfield Marine Sciences Centre.

To input new data:

1. Double click to open the appropriate form (Algal % Cover form etc.)
2. Along the bottom of the form you will see “Record:”

Date	31-Jul-07
Site	Wizard
Exposure	Exposed Sheltered
Transect #	29
Tidal Height	0.5 1.0 1.5 2.0 2.5 3.0 3.5

Species	Cover %
Acrosiphonia spp	0
Ahnfeltia pacifica	0
Alaria marginata	0
Anelopus japonicus	0
Chondracanthus exasperatus	0
Cladophora spp	2
Codium fragile	0
Copamania spp	0
Corallina vancouveriensis	0
Egregia menziesii	0
Encrusting coralline algae	0
Endocladia muricata	0
Enteromorpha intestinalis	0
Filamentous rhodophyta	0
Fucus spp	115
Gastroclonium coulteri	0
Halosaccion glandiforme	0
Hedophyllum sessile	0
Laminaria setchelli	0
Leathesia difformis	0
Mastocarpus papillatus	70
Mazzaella splendens	0
Microcladia borealis	0
Microcladia coulteri	0
Nemalion spp	0
Neorhodomela larix	0
Odonthalia floccosa	0
Pelvitopsis spp	0
Petrocelis middendorffii	0
Phyllospadix scouleri	0
Porphyra spp	0
Prasiola spp	0
Prionitis spp	0
Sargassum muticum	0
Scytosiphon lomentaria	0
Ulva spp	0
Bare rock	0

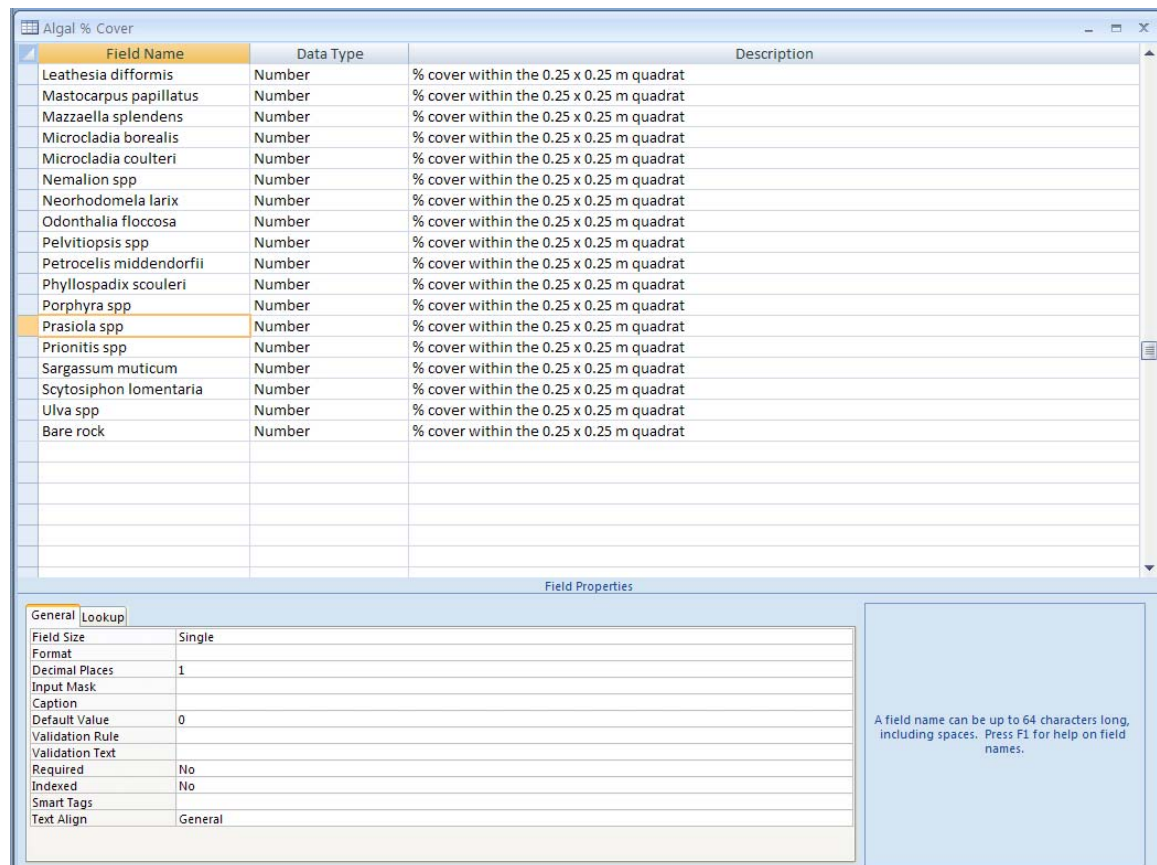
Record: 1080 of 1080 No Filter Search

3. Click on the button that says new blank record **OR** click to the last record and hit forward
4. This should give you a blank form which you can fill in with the new data
5. Click next when you are finished with that form and type in a new set of data
6. **Remember: Data entered into forms goes directly into the appropriate table**

To add a species to a table and form:

Occasionally a species will come up on your data sheets that were not seen in previous years. If this is the case you will need to first add the species to the table and then add it to the form.

1. Go to your tables page and double click the table you wish to add the species to
2. Go to design view
3. Insert a row in the appropriate alphabetical placement



Field Name	Data Type	Description
Leathesia difformis	Number	% cover within the 0.25 x 0.25 m quadrat
Mastocarpus papillatus	Number	% cover within the 0.25 x 0.25 m quadrat
Mazzaella splendens	Number	% cover within the 0.25 x 0.25 m quadrat
Microcladia borealis	Number	% cover within the 0.25 x 0.25 m quadrat
Microcladia coulteri	Number	% cover within the 0.25 x 0.25 m quadrat
Nemalion spp	Number	% cover within the 0.25 x 0.25 m quadrat
Neorhodomela larix	Number	% cover within the 0.25 x 0.25 m quadrat
Odonthalia floccosa	Number	% cover within the 0.25 x 0.25 m quadrat
Pelvitopsis spp	Number	% cover within the 0.25 x 0.25 m quadrat
Petrocelis middendorffii	Number	% cover within the 0.25 x 0.25 m quadrat
Phyllospadix scouleri	Number	% cover within the 0.25 x 0.25 m quadrat
Porphyra spp	Number	% cover within the 0.25 x 0.25 m quadrat
Prasiola spp	Number	% cover within the 0.25 x 0.25 m quadrat
Prionitis spp	Number	% cover within the 0.25 x 0.25 m quadrat
Sargassum muticum	Number	% cover within the 0.25 x 0.25 m quadrat
Scytosiphon lomentaria	Number	% cover within the 0.25 x 0.25 m quadrat
Ulva spp	Number	% cover within the 0.25 x 0.25 m quadrat
Bare rock	Number	% cover within the 0.25 x 0.25 m quadrat

Field Properties

General | Lookup

Field Size	Single
Format	
Decimal Places	1
Input Mask	
Caption	
Default Value	0
Validation Rule	
Validation Text	
Required	No
Indexed	No
Smart Tags	
Text Align	General

A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.

4. Down below is the Field Properties List. On your newly added species **MAKE SURE** that the Field Properties match the other species within that table. This makes the data all in one format.
5. Once you have added the species close the design view. It will ask you to save. Click Yes.
6. Go to the corresponding form for the table that you have just changed and open it
7. In design view you will need to bring up your field list and add the new species that you added to the table. **THIS IS IMPORTANT AS THE SPECIES IS NOT AUTOMATICALLY ADDED TO THE FORM WHEN IT IS ADDED TO THE TABLE.**

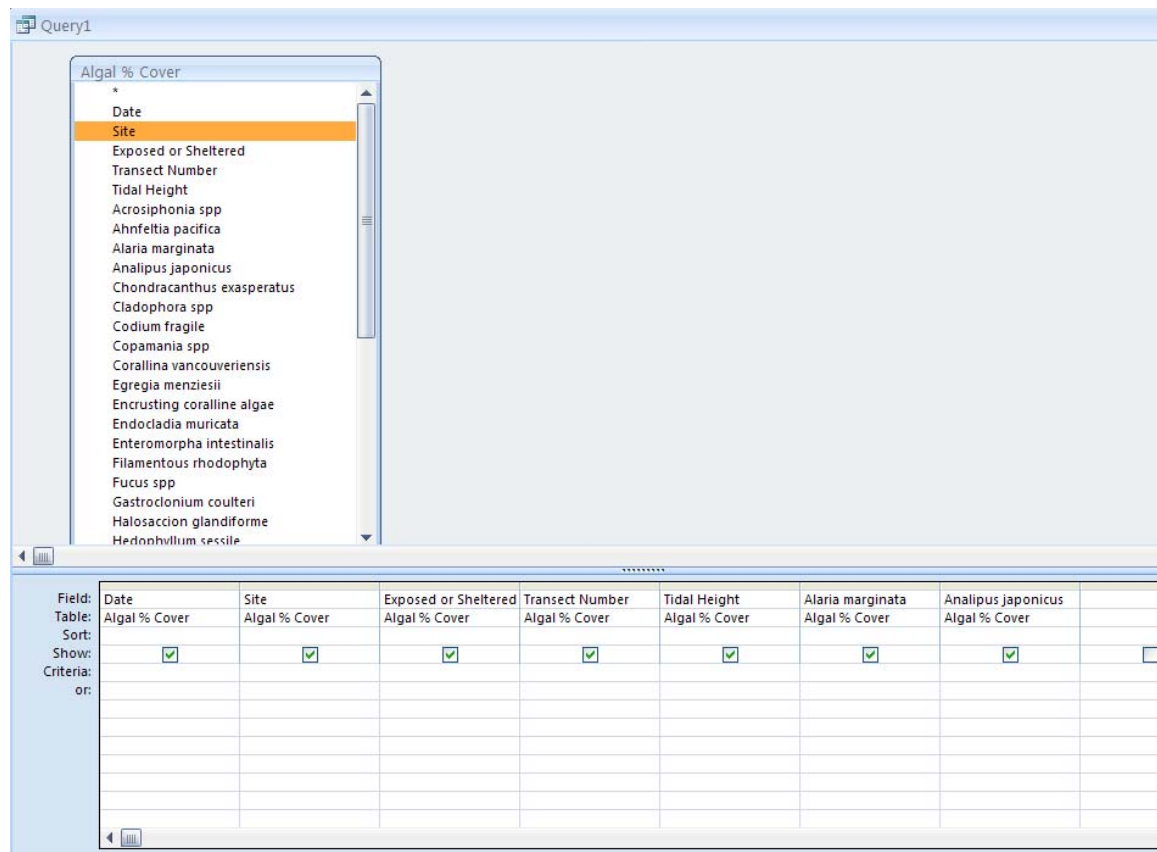
8. Format the table to be in alphabetical order
9. Continue entering data as before

Querying a subset of data:

ANY QUERY WITH “09” AT THE END IS OLD AND ONLY CONTAINS DATA UP TO ‘09

Querying is an important aspect of access as it allows you to get an almost instant subset of your data. This subset can be just specific species, just % covers over 50%, or many other possibilities. You can query tables or other queries if you need. For a full list of potential queries see the Access Help. Here is an example query:

1. Open up a new query in design view
2. Choose which table or other query you would like to query



3. Double click on the variables that you want to include in the query
4. The large exclamation point “!” will RUN your query
5. This will produce a table containing only the fields that you wish to be in your new query (in this case the 7 variable chosen above)
6. Within “Criteria” you can control which data from that variable you would like in your query (See Access help for common criteria)

Exporting between Access and Excel:

Data between the two programs are virtually interchangeable. Some data management is easier in Excel (especially if you are more comfortable with it). All of the data should, however, reside in the Access database and only be exported to Excel for further examination.

1. Make a new Excel file and save it to a known location
2. Close the Excel file (**This is important because the files won't export unless the Excel file is closed**)
3. In Access go to File → Export
4. Choose which table or query you would like to export and send it to the Excel file that you have just made
5. When you reopen the Excel file it should contain your Access table that you exported

Access can be “finicky” but is a great program for storing and analyzing large datasets such as this one. Issues may come up when entering data, but don't get discouraged. The HELP in Access can answer most questions.

If an issue cannot be resolved regarding this database feel free to contact the UP coordinator to get contact information for Josh Silberg (former student who helped build the database) or contact someone on the station with experience with Access.